SYNOPSIS
ON
CLOUD KITCHEN - PLATFORM FOR HEALTHY HOME - COOKED FOOD

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF
THE DEGREE OF

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1. ABSTRACT

With the increasing demand for healthy and homemade food, many individuals prefer nutritious meals over fast food. However, access to such food is limited, and many skilled home cooks, especially homemakers, lack a proper platform to sell their products.

This project proposes a Home-Based Cloud Kitchen Platform that connects home cooks with customers seeking healthy, organic, and homemade food. The system allows users to browse food items, place orders, and receive deliveries directly from nearby home kitchens.

The platform integrates a secure digital payment system that enables seamless transactions between buyers, sellers, and the platform. The payment module ensures proper distribution of earnings, where a portion is allocated to the seller, a delivery fee is assigned to the delivery partner, and a commission is retained by the platform.

Customers can also rate and review food items, ensuring quality and trust. The system is developed using Python and integrates database management and user-friendly interfaces to provide a seamless experience.

2. INTRODUCTION

Food delivery platforms have grown rapidly in recent years, but most of them focus on restaurants and commercial kitchens. These platforms often lack options for homemade, organic, and small-portion food items.

At the same time, many homemakers and small-scale food producers have the skills to prepare high-quality food but lack access to a digital marketplace. This creates a gap between supply and demand.

The proposed system aims to bridge this gap by providing a platform where home cooks can sell their food items directly to customers. The system supports small food items such as buttermilk, salads, and homemade snacks, making it flexible and inclusive.

Additionally, the system includes a secure and transparent payment mechanism that allows users to pay digitally while ensuring proper allocation of funds among all stakeholders involved in the transaction.

3.PROBLEM STATEMENT

Despite the availability of food delivery services, there is a lack of platforms dedicated to homemade and healthy food options.

Customers often struggle to find fresh, organic, and affordable homemade meals. At the same time, home cooks face challenges in reaching potential customers due to the absence of a structured system.

Another major challenge is the lack of secure and transparent payment systems that can efficiently manage transactions and fairly distribute payments among sellers, delivery partners, and the platform.

Therefore, there is a need for a platform that enables home-based food sellers to connect with customers while ensuring quality, convenience, accessibility, and secure financial transactions.

4. PROJECT OBJECTIVES

1.To develop a platform for connecting home cooks with customers

2. To provide healthy, homemade, and organic food options

3. To enable location-based food search and ordering

4. To implement secure login and authentication

5. To integrate a digital payment system for online transactions

6. To ensure fair distribution of payments among seller, delivery partner, and platform

7. To allow users to view product details including pricing and quantity

8. To provide ratings and reviews for quality assurance

9. To support real-time tracking of orders

5. DESIGN / METHODOLOGY

The system follows a structured approach consisting of multiple stages.

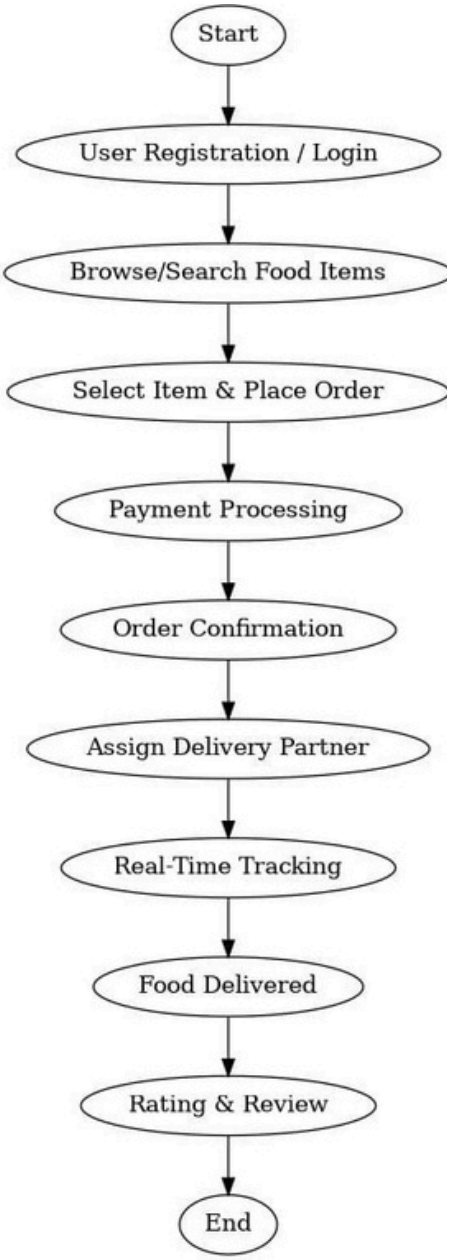
Initially, users (buyers and sellers) register and log in to the system. Sellers upload food items along with details such as price, quantity, availability time, and images.

The system uses location-based filtering to display nearby food options to users. Customers can search for food items using filters such as organic, vegetarian, and non-vegetarian categories.

Once a user places an order, the system processes the request and integrates a payment module. The total amount is divided into different components, including the seller's earnings, delivery charges, and platform commission.

The payment is processed using secure payment gateways, ensuring safe and reliable transactions. After successful payment, the order is confirmed and assigned to a delivery partner.

Real-time tracking is provided to monitor order status. The backend is developed using Python, where data is managed using databases, and the frontend ensures smooth interaction.



6. TOOLS AND TECHNOLOGIES

Software Technologies

1. Python – Core programming language used for backend development

2.Flask / Streamlit – Frameworks used to build the web application and user interface

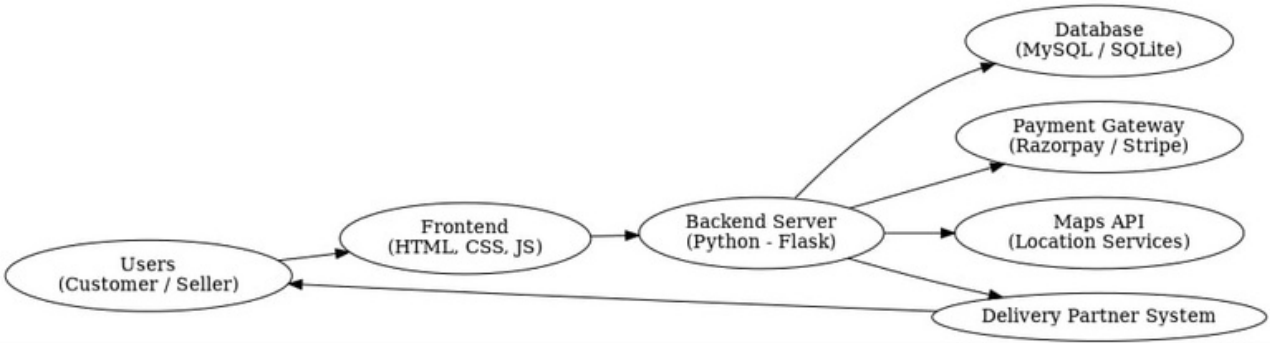
3.SQLite / MySQL – Database systems used to store user, product, and order data

4.HTML, CSS, JavaScript – Used for designing the frontend interface

5.Payment Gateway APIs (Razorpay / Stripe) – Used for secure online payment processing and transaction handling

6.Google Maps API – Used for location-based services and delivery tracking

7.Git & GitHub – Used for version control and project management



Hardware Technologies

1. Smartphone / Android Device – Used by customers and sellers to access the application

2. Computer / Laptop – Used for development, testing, and deployment of the system

3.Internet Connectivity (Wi-Fi / Mobile Data) – Required for real-time communication, ordering, and payments

4.GPS / Location Services – Used for tracking delivery and showing nearby food options

7. EXPECTED OUTCOME

1.Platform for Home Cooks and Customers
A fully functional system connecting home cooks with buyers
Mapped with Objective: “To develop a platform for connecting home cooks with customers”

2. Availability of Healthy Homemade Food
Users can access organic, homemade, and nutritious food easily
Mapped with Objective: “To provide healthy, homemade, and organic food options”

3. Efficient Search and Ordering System
Users can search food based on location, category, and preferences
Mapped with Objective: “To enable location-based food search and ordering”

4.Secure Payment and Transaction System
Implementation of a secure payment system with proper fund distribution
Mapped with Objectives:
“To integrate a digital payment system”
“To ensure fair distribution of payments”

5.Transparent Earnings Distribution
Seller, delivery partner, and platform receive their respective shares
Mapped with Objective: “To ensure fair distribution of payments”

6.Improved User Experience
Features like login, product details, ratings, and reviews enhance usability
Mapped with Objective: “To allow users to view product details and provide ratings”

7.Real-Time Order Tracking
Users can track their orders from placement to delivery
Mapped with Objective: “To support real-time tracking of orders”

8.Empowerment of Home-Based Sellers
Provides income opportunities for homemakers and small-scale cooks
Mapped with Objective: “To develop a platform for connecting home cooks with customers”

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Python Official Documentation
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